# ИДЗ №2 1 вариант

1. $\begin{cases} y''\cos y + y'^2 \sin y = y' \\ y(\ln(2-\sqrt{3})) = \frac{\pi}{6} \\ y'(\ln(2-\sqrt{3})) = \frac{1}{2} \end{cases}$

Вводим замену $z = y' \Rightarrow y'' = z_y' \cdot y_x' = z z'$

$z' z \cos y + z^2 \sin y = z \mid : z \neq 0$

($z = 0$ - решение, но не удовлетворяющее задаче Коши)

$z' \cos y + z \sin y = 1$

$z' + z \operatorname{tg} y = \frac{1}{\cos y}$

$\exp\left(\int \frac{\sin y}{\cos y} dy\right) = \exp\left(-\int \frac{d\cos y}{\cos y}\right) = \exp(-\ln \cos y) =$

$= \frac{1}{\cos y}$ - интегрирующий множитель

$\frac{z'}{\cos y} + \frac{z \sin y}{\cos^2 y} = \frac{1}{\cos^2 y} \Rightarrow \frac{z' \cos y + z \sin y}{\cos^2 y} = \frac{1}{\cos^2 y} \Rightarrow$

$\Rightarrow \left(\frac{z}{\cos y}\right)' y' = \frac{1}{\cos^2 y} \Rightarrow \frac{z}{\cos y} = \operatorname{tg} y + C$

$y' = \sin y + C \cos y \Rightarrow \frac{1}{2} = \frac{1}{2} + C \cdot \frac{\sqrt{3}}{2} \Rightarrow C = 0 \Rightarrow$

$\Rightarrow y' = \sin y \Rightarrow \frac{dy}{\sin y} = x + \hat{C} \Rightarrow \int \frac{\sin y}{\sin^2 y} dy = x + \hat{C}$

$-\int \frac{d\cos y}{1 - \cos^2 y} = x + \hat{C} \Rightarrow \int \frac{d\cos y}{\cos^2 y - 1} = x + \hat{C} \Rightarrow$

$\Rightarrow \frac{1}{2} \ln\left|\frac{\cos y - 1}{\cos y + 1}\right| = x + \hat{C}$

$\frac{1}{2} \ln\left|\frac{\frac{\sqrt{3}}{2} - 1}{\frac{\sqrt{3}}{2} + 1}\right| = \ln(2 - \sqrt{3}) + \hat{C}$

$\frac{1}{2} \ln\left|\frac{\sqrt{3} - 2}{\sqrt{3} + 2}\right| = \ln(2 - \sqrt{3}) + \hat{C}$

$\frac{1}{2} \ln|\sqrt{3} - 2| - \frac{1}{2} \ln|2 + \sqrt{3}| = \ln(2 - \sqrt{3}) + \hat{C}$

$-\frac{1}{2}\left(\ln|2 - \sqrt{3}| + \ln(2 + \sqrt{3})\right) = \hat{C}$

$-\frac{1}{2}(\ln(4 - 3)) = \hat{C} \Rightarrow \hat{C} = 0 \Rightarrow \boxed{2x = \ln\left|\frac{\cos y - 1}{\cos y + 1}\right|}$

2. $y'' - 2y' + y = 14t^{3/2}e^t$

Решим: $y'' - 2y' + y = 0$

$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow (\lambda - 1)^2 = 0 \Rightarrow \lambda_{1,2} = 1$

$y = C_1 e^t + C_2 t e^t$; $C_1 = C_1(t)$, $C_2 = C_2(t)$

$$\begin{cases} C_1' e^t + C_2' t e^t = 0 \\ C_1' e^t + C_2' e^t + C_2' t e^t = 14t^{3/2} e^t \end{cases}$$

$$\begin{cases} C_1' + C_2' t = 0 \\ C_1' + C_2' + t C_2' = 14t^{3/2} \end{cases}$$

$C_2' = 14t^{3/2} \Rightarrow C_2 = 14 \int t^{3/2} dt \Rightarrow C_2 = \frac{28}{5} t^{5/2} + \tilde{C}_2$

$C_1' = -14t^{5/2} \Rightarrow C_1 = -14 \int t^{5/2} dt \Rightarrow C_1 = -\frac{28}{7} t^{7/2} + \tilde{C}_1 =$

$= -4t^{7/2} + \tilde{C}_1$

$y = -4t^{7/2} e^t + \tilde{C}_1 e^t + \frac{28}{5} t^{7/2} e^t + \tilde{C}_2 t e^t =$

$= \frac{8}{5} t^{7/2} e^t + \tilde{C}_1 e^t + \tilde{C}_2 t e^t = y$

3. $y'' + 16y = \underbrace{\frac{1}{5}\sin 4x}_{f_1} + \underbrace{e^x \sin x}_{f_2}$; хар-ое уравнение:

$\lambda^2 + 16 = 0 \Rightarrow \lambda = \pm 4i$

• $y'' + 16y = f_1 = e^0 (0 \cdot \cos 4x + \frac{\sin 4x}{5})$

$a = 0$; $b = 4$; $Y_1(x) = x(C_1 \cos 4x + C_2 \sin 4x) = C_1 x \cos 4x + C_2 x \sin 4x$

$kp - mb = 1$

$Y_1' = C_1 \cos 4x - 4C_1 x \sin 4x + C_2 \sin 4x + 4C_2 x \cos 4x$

$Y_1'' = -4C_1 \sin 4x - 4C_1 \sin 4x - 16C_1 x \cos 4x + 4C_2 \cos 4x +$

$+ 4C_2 \cos 4x - 16C_2 x \sin 4x$

$-4C_1 \sin 4x - 4C_1 \sin 4x - 16C_1 x \cos 4x + 4C_2 \cos 4x +$

$+ 4C_2 \cos 4x - 16C_2 x \sin 4x + 16C_1 x \cos 4x + 16C_2 x \sin 4x =$

$= \frac{\sin 4x}{5} \Rightarrow -8C_1 = \frac{1}{5}$; $C_2 = 0 \Rightarrow C_1 = -\frac{1}{40} \Rightarrow$

$\Rightarrow Y_1(x) = -\frac{x}{40} \cos 4x$

• $y'' + 16y = e^x \sin x = e^{1\cdot x}(0\cdot\cos x + 1\cdot\sin x)$

$a=1;\ b=1;\ \alpha=0 \Rightarrow \psi_2(x) = e^x(C_1\cos x + C_2\sin x)$

$\psi_2'(x) = e^x(C_1\cos x + C_2\sin x) + e^x(-C_1\sin x + C_2\cos x)$

$\psi_2''(x) = e^x(C_1\cos x + C_2\sin x) + 2e^x(-C_1\sin x + C_2\cos x) +$
$+ e^x(-C_1\cos x - C_2\sin x)$

$e^x(C_1\cos x + C_2\sin x) + 2e^x(-C_1\sin x + C_2\cos x) +$
$+ e^x(-C_1\cos x - C_2\sin x) + 16e^x(C_1\cos x + C_2\sin x) = e^x\sin x$

$$\begin{cases} -2C_1 + 16C_2 = 1 \\ 16C_1 + 2C_2 = 0 \Rightarrow C_2 = -8C_1 \Rightarrow -2C_1 - 128C_1 = 1 \end{cases}$$

$C_1 = -\frac{1}{130}$

$C_2 = \frac{4}{65}$

$\psi_2(x) = e^x\left(-\frac{1}{130}\cos x + \frac{4}{65}\sin x\right)$

• $y'' + 16y = 0 \Rightarrow y = C_1\cos 4x + C_2\sin 4x \Rightarrow$

$\Rightarrow$ решение ур-ия: $y = C_1\cos 4x + C_2\sin 4x + \frac{x}{40}\cos 4x +$
$+ \frac{e^x}{130}(-\cos x + 8\sin x)$

4 $x^2y''\ln x - xy' + y = 0 \Rightarrow y'' - \frac{y'}{x\ln x} + \frac{y}{x^2\ln x} = 0$

$y = x$ – решение

$$\begin{vmatrix} x & y \\ 1 & y' \end{vmatrix} = C\exp\left(+\int\frac{dx}{x\ln x}\right) = C_1\exp(\ln|\ln x|) = C_1\ln x$$

$xy' - y = C_1\ln x \Rightarrow \frac{xy' - y}{x^2} = \frac{C_1\ln x}{x^2} \Rightarrow \left(\frac{y}{x}\right)' = \frac{C_1\ln x}{x^2}$

$\frac{y}{x} = C_1\int\frac{\ln x}{x^2}dx = \left|\begin{matrix} u = \ln x;\ du = \frac{dx}{x} \\ dv = \frac{dx}{x^2};\ v = -\frac{1}{x} \end{matrix}\right| =$

$= -C_1\frac{\ln x}{x} + C_1\int\frac{dx}{x^2} = -C_1\frac{\ln x}{x} - \frac{C_1}{x} + C_2$

$y = -C_1\ln x + C_2x - C_1$

5)

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \frac{1}{\cos t} \\ 0 \end{pmatrix}$$

Решаем $\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$$|A - \lambda E| = \begin{vmatrix} 1-\lambda & -1 \\ 2 & -1-\lambda \end{vmatrix} = (\lambda - 1)(1+\lambda) + 2 = \lambda^2 - 1 + 2 = \lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$$\begin{pmatrix} 1-i & -1 \\ 2 & -1-i \end{pmatrix} \sim \begin{pmatrix} 1-i & -1 \\ 1 & \frac{-1-i}{2} \end{pmatrix} \sim \begin{pmatrix} 1-i & -1 \\ 1-i & -1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow \vec{v}_1 = \begin{pmatrix} 1 \\ 1-i \end{pmatrix}; \quad \vec{v}_2 = \begin{pmatrix} 1 \\ 1+i \end{pmatrix}$$

$\lambda_1 = i \qquad \lambda_2 = -i$

$$\operatorname{Re}\left(e^{it}\begin{pmatrix} 1 \\ 1-i \end{pmatrix}\right) = \operatorname{Re}\begin{pmatrix} \cos t + i \sin t \\ (\cos t + i \sin t)(1-i) \end{pmatrix} =$$

$$= \operatorname{Re}\begin{pmatrix} \cos t + i \sin t \\ \cos t + \sin t + i \sin t - i \cos t \end{pmatrix} =$$

$$= \begin{pmatrix} \cos t \\ \cos t + \sin t \end{pmatrix}$$

$$\operatorname{Im}\left(e^{it}\begin{pmatrix} 1 \\ 1-i \end{pmatrix}\right) = \begin{pmatrix} \sin t \\ \sin t - \cos t \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} x = C_1 \cos t + C_2 \sin t \\ y = C_1 \cos t + C_1 \sin t + C_2 \sin t - C_2 \cos t \end{cases} \text{ — общее решение} \Rightarrow$$

однородное

$$\Rightarrow \dot{x} = C_1' \cos t - C_1 \sin t + C_2' \sin t + C_2 \cos t =$$

$$= C_1 \cos t + C_2 \sin t - C_1 \cos t - C_1 \sin t - C_2 \sin t + C_2 \cos t + \frac{1}{\cos t}$$

$$\dot{y} = C_1' \cos t - C_1 \sin t + C_1 \cos t + C_1' \sin t + C_2' \sin t + C_2 \cos t -$$

$$- C_2' \cos t + C_2 \sin t = 2C_1 \cos t + 2C_2 \sin t - C_1 \cos t - C_1 \sin t -$$

$$- C_2 \sin t + C_2 \cos t$$

$$\begin{cases} C_1' \cos t + C_2' \sin t = \frac{1}{\cos t} \\ C_1' \cos t + C_1' \sin t + C_2' \sin t - C_2' \cos t = 0 \end{cases}$$

$$\begin{cases} C_1' \cos t + C_2' \sin t = \frac{1}{\cos t} \\ C_1' \sin t - C_2' \cos t = -\frac{1}{\cos t} \end{cases}$$

$$\Delta = \begin{vmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{vmatrix} = -1$$

$$\Delta_1 = \begin{vmatrix} \frac{1}{\cos t} & \sin t \\ -\frac{1}{\cos t} & \cos t \end{vmatrix} = 1 + \operatorname{tg} t; \quad \Delta_2 = \begin{vmatrix} \cos t & \frac{1}{\cos t} \\ \sin t & -\frac{1}{\cos t} \end{vmatrix} = -1 - \operatorname{tg} t$$

$C_2' = 1 + \operatorname{tg} t \Rightarrow C_2 = t - \ln|\cos t| + \tilde{C}_2$

$C_1' = -1 - \operatorname{tg} t \Rightarrow C_1 = -t + \ln|\cos t| + \tilde{C}_1$

$$\begin{cases} x = (-t + \ln|\cos t| + \tilde{C}_1)\cos t + (t - \ln|\cos t| + \tilde{C}_2)\sin t \\ y = (-t + \ln|\cos t| + \tilde{C}_1)(\cos t + \sin t) + (t - \ln|\cos t| + \tilde{C}_2)(\sin t - \cos t) \end{cases}$$